Spheroid + ECM — archived sections

Split out of `spheroid_ecm.tex`. These sections describe the model as it stood before the external audit of § "External audit" below, and before junction myosin was re-keyed from length to tension and the membrane made overdamped. They are kept because the measurements in them are real and because the audit is the work list; they are not kept because the conclusions stand.

6. Status and the next operator

63 runs in `log/okuda_ECM`; `prototype/ecm/RUNS.md` records which are mutually comparable, since three quantities were redefined mid-campaign (notably $|J-1| \to \sigma_{\mathrm{vM}}$, $\sim 100\times$ in "strained fraction"). The coupling of §4 is one-way; `ecm_load_3d` is the staggered two-way scheme, implemented and uncalibrated. `cell_polarity` is absent—the epithelium has no simulated basal side, its cells being wedges from the centre and the monolayer a rendering convention—as are `cell_extrude_3d` and MMP degradation.

The clearest gap is in Eq. (**??**). With N fixed on a shell whose radius grows 3.6×, the areal density $N/4\pi R^2$ falls 13× and every rest length must creep to 3.6× its original span: what τ_r delivers is therefore *dilution with amnesia*, not remodelling, and it is why the surviving sheet sits permanently near 10% strain. A basement membrane does not thin thirteenfold as an acinus grows—the epithelium secretes laminin and collagen IV continuously and thickness is held roughly constant. The missing operator is `basement_membrane_secrete`, kind `structural`: insert particles where areal density has fallen below target and bond them in, the sheet's analogue of `divide_3d`. Plexus supports it directly through `Level.free_slots`/`spawn` with an oversized reservoir. Only then does "the membrane survived" mean what it means biologically—that the tissue did not outgrow the membrane it was building—rather than that its crosslinks were stretched and told to forget.

1 Two things the implementation gets wrong, stated plainly

The anchor sets where the sheet sits, and only a narrow band of it works. With critical damping the sheet stopped tearing, but it then sank *through* the epithelium: the gap to the recorded surface ran $+0.0040 \to -0.0101 \to -0.0255$, every particle inside by mid-run. Tracking lag does not explain it ($2v_a/\sqrt{k} = 0.0019$, sixteen times too small). Nor—and this corrects the obvious guess—does hoop tension: quartering k_{bond} from 2×10^5 to 5×10^4 at fixed k_{adh} moves the gap by under 5% (+0.00185/+0.00183/+0.00176) and leaves the strain alone. The reason is in Eq. (**??**)'s companion: remodelling fixes the steady-state strain at $\dot\epsilon\,\tau = 0.0027\times60 = 0.16$ (measured 0.151–0.160 in every run, at every stiffness), so rest lengths adapt and bond force never becomes large. What sets the gap is k_{adh} against a residual inward force; the untested candidate is the interstitial matrix pressing through the shared grid, which would be the one real matrix-on-membrane effect in the model.

The sweep is narrow. At 200 frames $k_{\mathrm{adh}} = 4\times10^4/6\times10^4/8\times10^4$ gives gaps $-0.0037/-0.0013/+0.0019$; 1×10^5 *reverses* to -0.069 and 2×10^5 tears 94% of the crosslinks. A stiffer spring cannot hold less well, so the reversal is a stability threshold rather than mechanics, and 8×10^4 is the usable value—at which 46% of particles are still inside. A membrane pinned at fixed angular positions on a surface that triples in radius is being asked to cover nine times the area with the same particles; the operator that would resolve this, and which the model does not have, is `basement_membrane_secrete`.

`activity` is degenerate with Λ. Actomyosin entered Eq. (**??**) as the single scalar Λ and now also as a per-junction variable, and the two do not double-count: myosin *multiplies* the line term, $\Lambda\sum_e m_e\ell_e$, and the recruitment setpoint of Eq. (**??**) is written against the *running mean* edge length $\langle\ell\rangle$, so

$$m_e^{ss} = a\Big(1+\beta\big(\ell_e/\langle\ell\rangle-1\big)\Big) \quad\Longrightarrow\quad \langle m^{ss}\rangle = a.$$

At $a=1$ the mean is pinned to one, Λ keeps the meaning it was calibrated with, and myosin is a pure zero-mean redistribution—which is why the ablation is bit-identical rather than merely close. But it follows that a, which scales

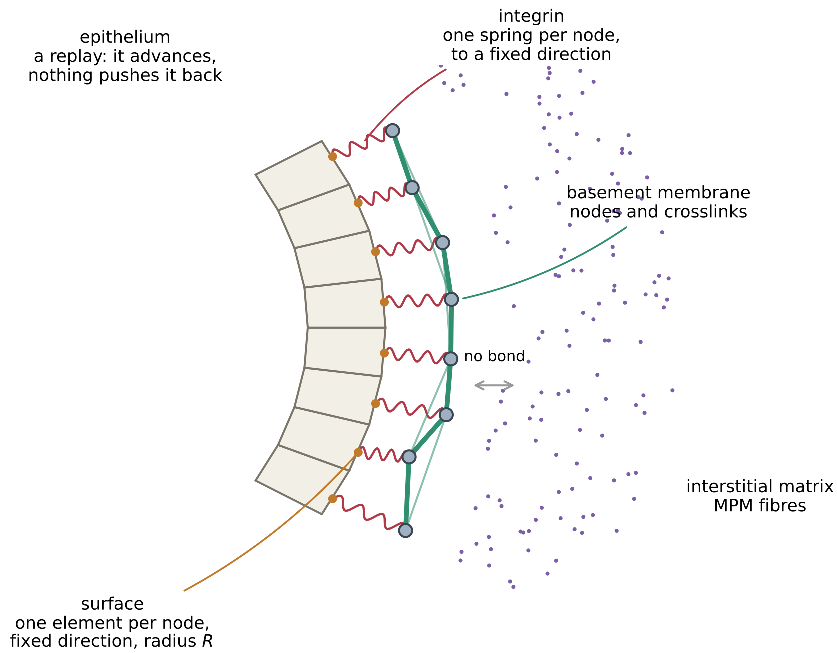

Figure 1: **What the membrane is made of.** Four objects, and the arrows between them are the whole model. The *epithelium* (cream wedges) is a recording: it advances on a prescribed schedule and nothing can push it back. Its outer boundary is the `surface` Level (orange), one element per membrane node, each holding a fixed direction $\hat{\mathbf{u}}_i$ and the radius $R(\hat{\mathbf{u}}_i, t)$ the tissue reached in that direction. The *basement membrane* is grey *nodes* joined by green *crosslinks*, sitting a distance δ outside. Each node is tied to its *own* surface element by one red *integrin* spring—one per node, to a fixed angular position, which is what makes it adhesion rather than contact. Outside is the interstitial matrix (purple), which the membrane never bonds to: they meet only through the shared grid, or in the spring-graph version, not at all.

every junction uniformly, is *not identifiable against* Λ: the monotone radius response over $a \in [0, 3]$ ($7.216 \to 6.138$) is reproduced by sweeping Λ and is a consistency check, not a result. The new parameters are β, τ and m_{new}.

The epithelial surface is not an entity. It is $R(\hat{\mathbf{u}}, t)$, a 32×64 angular table, read by four operators (`cell_to_ecm`, `cell_exclude_3d`, `integrin_adhesion`, `basement_membrane_seed`). Being a table and not a Level it carries no state and can receive no delta, which is the structural reason the adhesion of Eq. (**??**) is one-way. The Plexus2 form is a Level `surface`, one member per outer face with $(\mathbf{x}, \mathbf{v}, \hat{\mathbf{n}}, A)$, written each frame by a `broadcast` from the replayed mesh; the four operators then become ordinary `lateral` ones between two Levels, the angular lookup disappears, and—because a Level can be integrated—the integrin spring becomes mutual. A stronger version is available once the sheet is anchored, since the membrane particles are themselves a discretisation of that surface and could carry the contact directly; but runs without a membrane still need a contact path, so `surface` is the general answer and membrane-as-surface the specialisation.

2 The basement membrane, drawn out

A basement membrane is a thin sheet of protein that an epithelium lays down underneath itself and sits on. Two networks do the work: **laminin**, which cells grip, and **collagen IV**, which is crosslinked and carries the load; **nidogen** staples the two together and **perlecan** fills the mesh. We model the load-bearing one. So a node is not a molecule—it is a patch of sheet about a micron across—and a crosslink is not one collagen IV protomer but the stiffness of all of them across that patch.

Three rules, and everything in Fig. 2 follows from them.

1. The crosslinks make it a sheet. Every pair of neighbouring nodes is a spring with its own rest length ℓ^0_{ij}, and its *elongation* is what the colours in Fig. 2 show:

$$\epsilon_{ij} = \frac{L_{ij} - \ell^0_{ij}}{\ell^0_{ij}}, \qquad \mathbf{f}_{ij} = k\,(L_{ij} - \ell^0_{ij})\,\hat{\mathbf{d}}_{ij}, \qquad \text{bond dies if } \epsilon_{ij} > \epsilon_{\text{break}}.$$

Without them the nodes are dust: independent points that cannot tear, cannot fragment, and cannot be measured for either.

2. Integrin pins each node to a *direction*, not to the nearest surface. The anchor is

$$\mathbf{a}_i = \mathbf{c} + \hat{\mathbf{u}}_i\big(R(\hat{\mathbf{u}}_i, t) + \delta\big),$$

with $\hat{\mathbf{u}}_i$ frozen when the node is laid down. Anchoring to the *nearest* point instead would let a node slide freely along the surface—that is contact, not adhesion—and the sheet would never stretch at all. Run 73 is that experiment run on purpose, and its mean elongation is the lowest in the series outside the slack cases.

3. Remodelling lets the sheet forget. A crosslink's rest length creeps toward its current length on a timescale τ, $\dot{\ell}^0 = (L - \ell^0)/\tau$, which gives the relation that governs this whole series: a sheet stretched at rate $\dot{\varepsilon}$ settles at

$$\boxed{\langle \epsilon \rangle \simeq \dot{\varepsilon}\,\tau}$$

independent of stiffness and of supply. It is why k can be swept over three orders and change nothing, and why τ is the parameter that decides whether the membrane ever approaches failure.

4. Secretion adds material. A surface growing from R_0 to R_1 needs $(R_1/R_0)^2$ times as many nodes to stay the same thickness. Deposition is a fixed fraction ρ of existing material per frame, so there is a threshold, predictable before any run:

$$\rho^* = \frac{2\ln(R_1/R_0)}{N_{\text{frames}}} = \frac{\ln 12.77}{400} = 0.0064 \text{ per frame},$$

measured at 0.006.

How it is built in Plexus2

The hierarchy is four sets and one field. `cells` and `junction` are the epithelium; `surface` is its boundary; `basement_membrane` is the sheet; `interstitial_ecm` is the matrix; and `mpm_grid` is the field every particle body scatters into, which is how bodies that never touch each other exchange momentum.

The membrane's operators divide by what they do to the state, which is the distinction Plexus2 draws: `seed_basement_membrane` and `basement_membrane_secrete` are **structural** (they change how much material exists); `basement_membrane_bond` and `integrin_adhesion` are **dynamics** (they return a force the engine integrates); `basement_membrane_remodel` is **field**-like (it rewrites rest lengths in place); and `basement_membrane_bond_break` is **rewire**, because fragmentation *is* a change of the edge set and is registered as one.

Two implementation choices are worth naming because neither is forced by the biology. First, per-bond state is keyed by the *unordered pair of node indices* rather than stored by position in an array, so division, T1 and death permute the mesh freely and no topology operator needs to know that crosslinks exist. Second, the membrane no longer uses MPM: nothing about the sheet came from the continuum solver, and dropping it costs exactly one thing—momentum exchange with the matrix, measured as an ECM stress falling from 7.5 to 0.006. That coupling has to come back as an explicit operator (`basement_to_ecm`), which resolves at the sheet's own length scale instead of the grid's.

3 The junction network, drawn out

The rules

In the original vertex model, contractility is one number for the whole tissue: every junction pulls with the same line tension Λ. We keep Λ and give each junction a dimensionless multiplier m_e on top of it, so the line term of Eq. (**??**) becomes

$$\Lambda \sum_e \ell_e \;\longrightarrow\; \Lambda \sum_e m_e\,\ell_e.$$

1. Myosin chases a setpoint that depends on length. A junction longer than the current average recruits more; shorter recruits less. Written against the *running* mean $\langle \ell \rangle$,

$$m_e^{ss} = a\Big(1 + \beta\big(\ell_e/\langle \ell \rangle - 1\big)\Big), \qquad \dot{m}_e = \big(m_e^{ss} - m_e\big)/\tau_m.$$

Because the reference is the running mean, $\langle m^{ss} \rangle = a$ exactly: the average is pinned and myosin is a pure *redistribution*. At $a = 1$ nothing is double-counted with Λ, which is why the ablation is bit-identical rather than merely close.

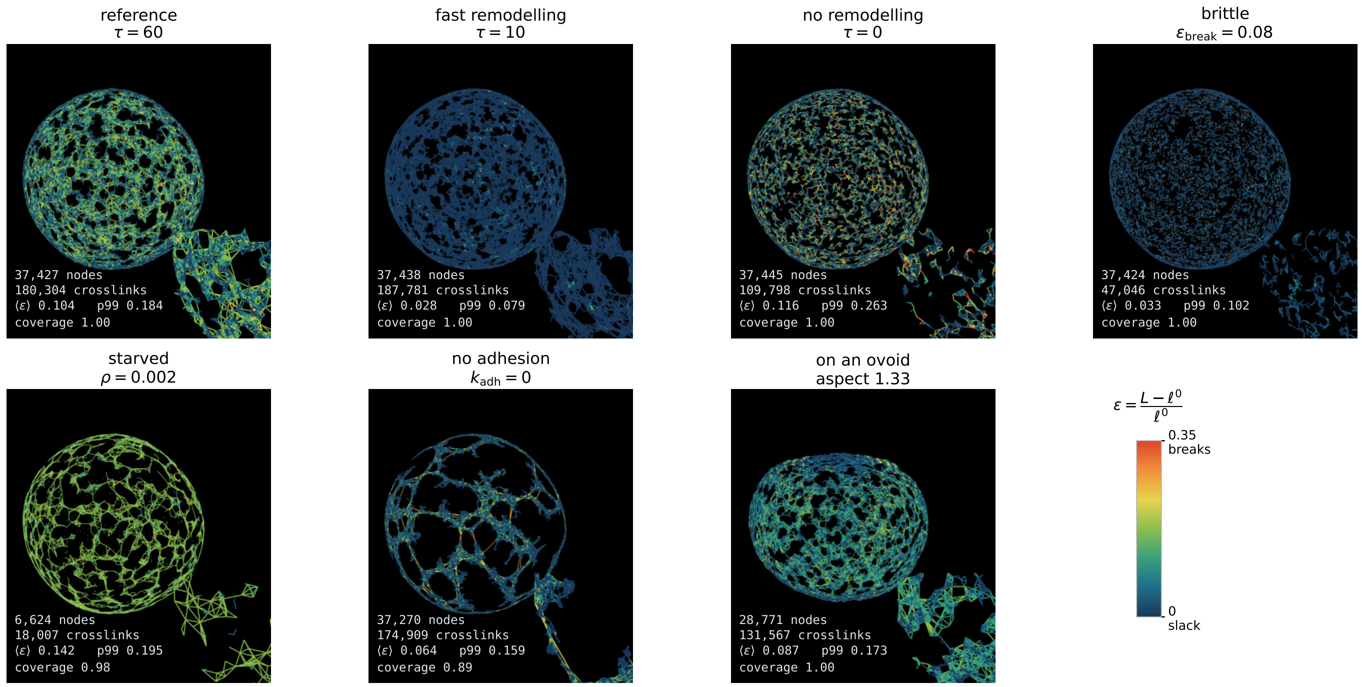

Figure 2: **Seven configurations, same tissue, end state.** Only the membrane is drawn: nodes grey, crosslinks coloured by elongation with full scale at the break threshold. Reading them against Eq. (boxed): **71** ($\tau = 10$) forgets stretch as fast as it arrives and is uniformly slack—the whole sheet is dark blue at $\langle \epsilon \rangle = 0.028$. **69** ($\tau = 60$) is the same network (180,304 against 187,781 crosslinks, both fully covering) under four times the tension, 0.104, and reads teal-green. **70** ($\tau = 0$, remodelling removed) cannot forget at all: tension accumulates, p99 reaches 0.263 against a threshold of 0.35, and 39% of the crosslinks have already gone (109,798). **74** lowers the threshold to 0.08 instead—the mechanics are untouched, but now the sheet is above it, and 74% of the crosslinks break (47,046); the survivors are the slack ones, which is why its mean elongation reads *lower* than the reference at 0.033. **72** is starved below ρ^*: 6,624 nodes instead of 37,427, a sheet stretched thin rather than torn. **73** removes the anchor and the sheet slides instead of stretching—and is the only one that loses coverage, 0.895, because nothing holds it in place. **75** is the same membrane on an ovoid (1.33), where the equator carries 33% more elongation than the poles.

2. And therefore a is not a new parameter, but β is. Since a multiplies every junction equally, it does exactly what scaling Λ does; the two cannot be told apart from any observation. β can: it changes the *disorder* of junction lengths at constant tissue size. Sweeping a over $[0.6, 1.4]$ traces a line in (radius, junction-length CV) of slope $+0.103$; β moves perpendicular to it, $\Delta R = -0.02$ (0.2%, nothing) against $\Delta \mathrm{CV} = -0.099$, where that line predicts -0.002. Measured:

	a	R	CV of ℓ_e
reference	1.0	10.94	0.321
$\beta = 2$	1.0	10.95	**0.249**
$\beta = 4$	1.0	10.92	**0.222**
blebbistatin-ish	0.6	11.12	0.350
hypercontractile	1.4	10.73	0.310

$\beta = 4$ reaches a CV no value of a in range approaches, at a radius that has not moved.

3. A junction born from a division inherits. Not every edge that appears is new. A division inserts a vertex into each of two edges and joins them, so it makes two *kinds*: a split half of a cut junction, which has a myosin history, and the daughter–daughter interface, which does not. Measured over 251 frames: of 13,750 edges that had no record, 9,920 (72%) were split halves and 9,760 of those (98.4%) recovered their parent. Only the remaining 28% take m_{new}.

4. τ_m was chosen, not measured. Stated plainly because it is the weakest number in this section: $\tau_m = 20$ frames is a default. It is swept nowhere except a determinism check. What it sets is how far myosin lags the length changes driving it, and the analogous relation for the membrane ($\langle \epsilon \rangle \simeq \dot{\varepsilon} \tau$) was measurable, so this one should be too.

Figure 3: **What a junction is, and the one loop that makes it interesting.** Left: two cells share an edge, and that edge *is* the junction—it belongs to neither cell, which is why its state is keyed by the unordered pair of vertices (i, j) rather than stored on a cell. Myosin m_e sits on the shared edge and sets its tension. Right: the feedback. A junction longer than average recruits more myosin (β), which pulls harder, which shortens it. That loop is the only thing here that Okuda's model cannot already do.

How it is built in Plexus2

One operator, `junction_myosin`, of kind **structural**: it does not return a force, it rewrites a per-edge quantity that the mechanics then reads in the same frame. It is scheduled immediately before `shape_energy_3d` for that reason—placed after, every relaxation would use the previous frame's myosin.

The design claim worth stating is what is *not* there. Junction is not a Level. Per-edge state is keyed by the unordered vertex pair, so `divide_3d`, `reconnect_t1_3d` and any future death operator permute, lengthen and shorten the half-edge arrays freely: an edge that still exists finds its own value, one that is new finds its parent or a default, one that is gone is simply never asked. None of the three topology operators knows that myosin exists, and adding the next per-edge state will not require editing them either. Promoting junction to a Level would buy per-edge provisioning and cost a reconciliation operator for every topology change.

4 External review, August 2026 — and what to do about it

An external reviewer with a cell-biology and biophysics background audited the code and this note. The findings below are theirs; the three marked **[verified]** I re-checked independently against the code and the run archive before recording them. Citations are as supplied and are *not* all independently confirmed; those I could not check are marked **[unchecked]**.

Verdict

The engineering is sound and self-critical; the biology is thinner than the prose implies, and three headline results are properties of the code rather than of the model.

The three most serious findings

1. The growth gate cannot see stress magnitude. [verified] The gate compares P/p_{99} to $p_{\mathrm{half}} = \mathrm{median}(P_{\mathrm{late}})/p_{99}$, so it evaluates $f(P/\mathrm{median}\,P)$: *multiplying the entire pressure field by any constant leaves it unchanged.* Every "the amount does not matter, only the pattern" control in this campaign is therefore an identity of the normalisation, not a finding. Compounding this, `smooth_phi_deg`$= 360$ forces the map axisymmetric before any cell divides, so an oblate spheroid is the only shape available. The reviewer notes that Hill $n = 6$ also sits beyond the one calibrated fit in the literature: Van Liedekerke et al. 2019 fit this exact form to Helmlinger/Montel data, obtained $n \in [1, 2]$, and explicitly reject $n \geq 4$. **[unchecked]** They further note our own primary citation, Helmlinger 1997, reports suppression accompanied by *decreased apoptosis with no significant change in proliferation.*

2. The membrane series is coupled to nothing, and its adjudicating metric is `NaN`. [verified] Runs 69–74 return *byte-identical* matrix metrics (`strained_frac_end` = 0.9635714285714285, identical `front_r95` arrays). In `membrane_impl="graph"` the sheet's MPM operators are stripped, so it exchanges no momentum with the matrix: the seven configurations are a sheet driven by a prescribed boundary, affecting nothing. Separately, `lcc_end` is `NaN` in every race run, because the component fraction is computed only every 40 frames and 403 is not a multiple of 40 — so the quantity §5 claims is reported "instead of the bond count" has never been reported. Worst, run 74's mean bond degree is $z = 2.51$ **[verified]**, below both rigidity ($z \approx 4$) and connectivity percolation, while runs 69–73, 75 sit at 5.4–10.0: that sheet most likely has no mechanical integrity, and Fig. 2's reading of it as "the survivors are the slack ones" is probably wrong.

3. Both new levels use the wrong state variable or the wrong integrator. Junction myosin is driven by *length*, not tension, and the docstring's identification of the two is false in this energy. The literature has myosin recruited by tension or strain rate, giving a *positive* feedback that drives T1; ours is a negative feedback on length that homogenises junctions. Worse, in the biologically correct fast-myosin limit the operator is analytically an edge spring: substituting $m_e = a(1 + \beta(\ell_e/L - 1))$ gives $\Lambda a \sum_e [(1-\beta)\ell_e + \beta\ell_e^2/L]$, a line tension plus a harmonic spring of stiffness $2\Lambda a\beta/L$. Reducing the CV of edge lengths is exactly what that does, so our one non-degenerate myosin result is guaranteed by construction. The reviewer also shows the `clamp_min(0)` breaks the $\langle m \rangle = a$ argument at $\beta = 4$ (estimated +16%), which is the row the table rests on. Separately, the membrane is integrated inertially with unit mass, so the whole critical-damping / tracking-lag / sinking narrative of §7 is the phenomenology of an inertia that should not exist at $\mathrm{Re} \sim 10^{-10}$.

Other findings worth keeping

• ρ^* is an identity of `basement_membrane_secrete`, not a prediction — and it was computed from the *ungated* tissue ($R_1/R_0 = 3.57$) while the sweep ran on the gated one, for which $\rho^* = 0.0055$, not 0.0064. • "Thinning, not rupture" is imposed: there is no thickness variable, and remodelling puts rupture out of reach by construction. The literature has BMs opening by *focal, protease-mediated perforation*, and being maintained or stiffening during growth. **[unchecked]** • Our integrin anchors to the collagen IV network; integrins bind *laminin*, and *Drosophila* has no collagen-binding integrin at all. • Break strain 0.35 is below every tissue-level BM failure measurement cited. • $\tau_m = 20$ frames is $\sim$3.5 h under the only sensible time anchor, against a junctional myosin FRAP $t_{1/2}$ of $\sim$6 s. • Our citation "Ku & Bilder (2023) *Dev. Cell* 58:522" is wrong and must be corrected or removed. • Our oblateness estimator returns 1.022 on a perfect sphere, so several reported effects sit at its own null.

The work list, in order

1. **Fix `lcc` and re-report 69–77 by connectivity.** Compute the component fraction on the final frame unconditionally, and record mean degree z alongside every bond count. No new simulation — the trajectories are on disk. This settles run 74 and decides "thinning vs fragmented".
2. **Three tissue passes that decide the ovoid:** `gate_hill`= 1 and = 2 at fixed floor, and one run at `smooth_phi_deg`= 90 so axisymmetry is not imposed. Then quote the estimator's sphere null and rewrite the "amount does not matter" controls as the normalisation identity they are.
3. **Give junction myosin the per-edge tension** (available from `shape_energy_3d`'s gradient) or the strain rate, report $\langle m \rangle$ against a, and measure something non-tautological — T1 rate or hexagonal fraction against a Λ-matched control.
4. **Make the epithelium feel the sheet**, for runs 70 and 74 only: they are the two configurations carrying real tension, and a local hernia through a fragmented patch is the one phenomenon a basement-membrane model exists to produce.
5. **Switch the membrane to overdamped integration**, which deletes §7 rather than fixing it.

5 External audit, and the work list it produces

An external reviewer with a cell-biology and biophysics background was asked to audit this prototype adversarially: to find where the biology is wrong, where a simplification changes a conclusion, and where a reported result is a property of the code rather than of the model. What follows is that audit, condensed. I have marked **[verified]** against every claim I checked independently before writing it down; those checks are cheap and are listed so the reader can repeat them.

Findings I confirmed

The growth gate cannot see stress magnitude. [verified] The gate reads press/$p_{1/2}$ where the map is normalised as P/P_{99} and $p_{1/2}$ is the late-run median in those same units. The normalisation cancels: the operator is a function of $P/\text{median}(P)$ alone, so multiplying the entire pressure field by any constant—stiffer matrix, denser matrix, larger k_{contact}—leaves the gate unchanged. The comment in `load_ops.py` states that the normalisation was introduced so that $p_{1/2}$ would mean the same thing whatever the matrix stiffness. That is an invariance built on purpose; §4's "five amplifications each raised the stress and lowered the shape" and "the shape comes from the pattern, not the amount" then report the null of a sweep the code guarantees to be null. Compounding it, `smooth_phi_deg`= 360 makes the map axisymmetric before any cell divides, so the output is constrained to be a spheroid of revolution, and $p_{1/2}$ is calibrated on the last quarter of the same run it gates from frame 0.

The Hill exponent is outside the range the one calibrated fit allows. Van Liedekerke et al. (2019, *PLoS Comput Biol* 15:e1006273) fit this exact functional form to the Helmlinger/Montel/Alessandri data and obtain $n \in [1, 2]$, stating that $n \geq 4$ produces a flattening of residual growth that is not observed. We use $n = 6$, and the only sweep that exists (4/6/8) spans the flat top of the switch and co-varies `floor` with `hill`, confounding the two. Integrating the gate on our own recorded map, the pole/equator contrast falls only about 16% at $n = 1$, so the ovoid may well survive—but $n = 1$ and $n = 2$ have never been run on the headline configuration.

Helmlinger measured the opposite of what we cite it for. Our primary citation for pressure-suppressed proliferation states in its abstract that stress-induced inhibition of plateau-phase spheroids "was accompanied by decreased apoptosis with *no significant changes in proliferation*". Delarue et al. (2014) locate the causal variable in cell *volume* acting through p27$^{\text{Kip1}}$, two abstractions from a Hill function of stress; and the sign is not universal (Aragona 2013, Gudipaty 2017 have proliferation permitted where stretch is high).

The oblateness estimator's own null is 1.020. [verified] On 200,000 uniformly sampled points of a perfect sphere, $r_{\text{eq}}/r_{\text{ax}}$ taken as ratio of 98th percentiles reads **1.020**, while the RMS form reads 1.000. So the reference tissue's "aspect 1.021" is exactly round, and every effect in the campaign below $\approx$1.05—including the myosin runs at 1.016–1.056—sits at or under one estimator's null. Two estimators are in use and neither null was quoted. This is a fourth quantity redefined mid-campaign and belongs in `DEFECTS.md`.

The membrane series is a sheet coupled to nothing. [verified] Runs 69–74 return *byte-identical* interstitial-matrix metrics—`strained_frac_end` = 0.9635714285714285, `max_disp` = 0.20738065242767334, `front_reaches_wall` = 235—to full float precision across all six. In `membrane_impl="graph"` the membrane's MPM operators are stripped, so the sheet exchanges no momentum with the matrix at any resolution. The §5 caveat about *dx* failing to resolve a 0.002 sheet is moot for these runs: there is no coupling to under-resolve.

The fragmentation metric has never returned a number. [verified] `lcc_end` reads `bt[-1,3]`, but the component fraction is only computed when the frame index is a multiple of `components_every`= 40 and is `NaN` otherwise. With 403 frames it is `NaN` in all 27 race runs and is not recorded at all for 69–77. §5 asserts that what is reported is the largest connected component rather than the bond count. It has never been reported.

Run 74 is probably not a sheet. [verified] 47,046 bonds on 37,424 live nodes is a mean degree of **2.51**, against $z \approx 4$ for central-force rigidity percolation in 2D and $z \approx 4.5$ for connectivity percolation. Figure 2 reads that panel as "the survivors are the slack ones"; the more likely reading is that the network has lost mechanical integrity altogether. `coverage`= 1.0 does not contradict this—it counts occupied angular bins, not connectivity.

$\langle\epsilon\rangle \simeq \dot{\varepsilon}\tau$ fails at the τ we use. With $\dot{\varepsilon} = 0.00314$/frame: $\tau = 10$ predicts 0.031 against 0.028 measured (good); $\tau = 60$ predicts 0.188 against **0.104** measured, 44% low. The values quoted as verification (0.151–0.160) come from the *non-secreting* runs; in the secreting series most of the final material was deposited unstrained, which dilutes the mean. The boxed law is presented as holding across the series and does not.

The secretion threshold is an identity, not a prediction. ρ^* follows in three lines from the operator's own setpoint $N_{\text{want}} = n_0(R/R_0)^2$. Deriving it and then measuring it is a unit test presented as a falsifiable prediction—and the number quoted uses the *ungated* tissue ($R_1/R_0 = 3.57$) while the sweep runs on the gated one (≈ 3.0, giving $\rho^* = 0.0055$), so the agreement is fortuitous at the sweep's resolution.

Junction myosin is keyed to the wrong variable

The setpoint uses $\ell_e/\langle\ell\rangle$: the state variable is *length*, not tension, and the docstring's identification of the two is false in this energy, where edge tension is $\Lambda m_e + \sum_f 2K_P(P_f - P_f^0)$ and length does not appear directly. The sign is also inverted relative to the papers we cite: Bertet et al. (2004) report myosin enriched in *disassembling* junctions, and Fernandez-Gonzalez et al. (2009) find more myosin on linked edges *regardless of edge length.* The modelling literature keys on tension or strain rate and never on length (Sknepnek et al. 2023 state verbatim that their active force "does not include the junction length"). So our feedback is a *stabiliser*—longer → more myosin → shorter, homogenising—where the biology is a *destabiliser* that generates T1s.

Worse for the identifiability argument of §9: in the $\tau_m \to 0$ limit, which is the biologically correct one, substituting the setpoint gives

$$E_{\text{line}} = \Lambda a \sum_e \Big[(1-\beta)\,\ell_e + \beta\,\ell_e^2/\langle\ell\rangle\Big],$$

a line tension plus *a harmonic edge spring of zero rest length.* Lowering the coefficient of variation of edge lengths is the defining property of such a spring. The measurement that β lowers CV at constant radius is therefore tautological: any length-restoring feedback produces it, so it cannot discriminate a correct law from an incorrect one. The observable that would is the **T1 rate**, which the two laws move in opposite directions and which we have never measured.

And $\langle m^{ss}\rangle = a$ holds only without the `clamp_min(0)`. At $\beta = 4$ with edge-length CV ≈ 0.32 the setpoint clamps for any edge shorter than $0.75\langle\ell\rangle$, biasing the mean up by an estimated 16%—so $\beta = 4$ also raises effective Λ, and "R did not move, therefore β is a pure redistribution" is not established. `MYOSIN_TRACE` already records `myo.mean()`, so this is checkable from logs on disk without rerunning anything.

Separately, $\tau_m = 20$ frames is not merely unmeasured but wrong by orders of magnitude: under the time anchor that makes τ_r defensible (≈ 10 min/frame), $\tau_m \approx 3.5$ h against a junctional myosin FRAP half-time of ≈ 6 s (Kasza et al. 2014).

Two more structural objections

The membrane is integrated inertially, at Re $\sim 10^{-10}$. `emit: acceleration` gives $v \mathrel{+}= dt\,a$; $x \mathrel{+}= dt\,v$ with unit mass, where the correct equation of motion is $\gamma\dot{\mathbf{x}} = \mathbf{F}$. The entire narrative of §5 and §7—the undamped spring oscillating about a moving anchor, critical damping $c = 2\sqrt{k}$, the tracking lag, the sheet sinking through the epithelium, the stability reversal above $k_{\text{adh}} = 10^5$—is the phenomenology of an artificial inertia. In the overdamped limit none of it exists. Those were reported as calibration facts found by tests rather than by inspection; they are numerical facts about a term that should not be in the model.

Integrin binds laminin, not collagen IV. Integrins $\alpha3\beta1$, $\alpha6\beta4$ and $\alpha6\beta1$ bind laminin; the collagen IV receptor is $\alpha1\beta1$, and *Drosophila* has no collagen-binding integrin at all. So the model couples cells to the membrane through a ligand its reference organism's integrins do not bind, while omitting the network they do. Junctional epidermolysis bullosa—laminin-332 *or* $\alpha6\beta4$ mutations—is the genetic proof that the link runs through laminin. Töpfer et al. (2022), which we cite, also finds *perlecan* contributes greatly to elongation, contradicting our description of it as filler. And a central-force network has no bending rigidity, which for a 100–300 nm sheet is arguably the dominant mode.

Also: `break_strain`= 0.35 sits below every tissue-level failure measurement found (lens capsule 108%, declining to ≈40% with age; *C. elegans* spermatheca stretching ≈70% and recoiling hundreds of times). And the literature does not support "thinning": basement membranes open by *regulated focal perforation*, and where imaged during growth they are maintained or thickened.

Citations to fix

"Ku & Bilder (2023) *Dev. Cell* 58:522" in `membrane_ops.py` does not exist; the real paper is Ku, Harris & Bilder 2023, *Dev Cell* 58(3):211–223, and it is about polar cells secreting an MMP, not basement-membrane deposition. "Eschenbruch et al. (2021) *Cells* 10:1979" could not be verified. Okuda 2018, Hu 2018, Töpfer 2022 and Villeneuve 2024 check out.

The work list, in the order the audit ranks it

1. **Fix `lcc` and re-report by connectivity.** One line, no new simulation: compute the component fraction unconditionally on the final frame, and add mean degree $z = 2n_{\text{bonds}}/n_{\text{alive}}$ to the reported dicts with the $z \approx 4$ rigidity threshold drawn on Figure 2. This settles run 74 and either validates or kills "thinning, not rupture", from trajectories already on disk.

2. **Three tissue passes that decide the ovoid.** $n = 1$ and $n = 2$ at fixed floor—the dose-response the literature supports—and one run at `smooth_phi_deg`$= 90$ to test whether the shape survives when axisymmetry is not imposed. Then quote the estimator null and rewrite the "amount does not matter" controls as the consequence of the normalisation that they are.

3. **Give junction myosin the right input.** Replace $\ell_e/\langle\ell\rangle$ with the per-edge tension already available from `shape_energy_3d`'s gradient, or with the strain rate $d\ln\ell_e/dt$. Print $\langle m\rangle$ to test the clamp bias from existing logs. Then report **T1 rate** and hexagonal fraction against a Λ-matched control—the observables that discriminate a homogenising length feedback from a destabilising tension one.

Standing large item: make the epithelium feel the sheet, for runs 70 and 74 only—the two configurations carrying real tension, and the two where a local hernia through a fragmented patch could finally appear.

Where the work stands after this

Two things survive intact. The software design—operator kinds, topological keying of per-edge state, the negative results recorded rather than buried—is not what the audit attacks. And the combination of a discrete crosslinked membrane, secretion-coupled membrane growth, and a 3D growing, dividing epithelium appears to be genuinely unpublished; the reviewer could find no model with all three. That novelty is conditional on the coupling becoming two-way and the fragmentation metric working.

What does not survive in its current form: the ovoid as a stress-driven result, the membrane failure-mode dichotomy, and the claim that per-junction myosin expresses a tension feedback.